

Syed Adil Hussain Shah, PhD

Artificial Intelligence Researcher

Department of Research and Development (R&D),

GPI S.p.A., Trento, Italy

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PROFESSIONAL LINK:

BIOGRAPHY:

AI Researcher at GPI SpA and Marie Skłodowska-Curie Actions Fellow within the European Union's Horizon 2020 programme, specialising in AI, Computer Vision, Biomedical Imaging, NLP, and Explainable AI. Author of 20+ peer-reviewed publications, with experience in international research collaboration, student supervision, applied AI development, and healthcare product design. Passionate about translating AI research into reliable, interpretable solutions for real-world clinical challenges.

EMPLOYMENT EXPERIENCE:

Artificial Intelligence Researcher

GPI SpA, Trento, Italy Aug 2022 - Present

Technical Expertise: AI for Digital Health, Voice Biomarkers, Clinical Decision Support, Multimodal AI, Explainable AI, and Healthcare Innovation

Contribute to multiple European and national AI healthcare projects involving hospitals, universities, research centres, and industry partners.

Develop AI/ML pipelines for digital health applications, including biomedical data analysis, voice biomarkers, multimodal data integration, and clinical decision support.

Lead project coordination activities using Jira, including task planning, documentation, milestone tracking, and technical reporting.

Support EU grant proposal writing, proof-of-concept design, prototype planning, and research-to-product translation for healthcare technologies.

Research Assistant

COMSATS University, Islamabad, Wah Campus, Pakistan Aug 2021- Jun 2022

Technical Expertise: Medical Image Analysis, Computer Vision, Deep Learning, and Disease Identification

Conducted research in image processing and computer vision for medical image analysis.

Developed deep learning approaches for medical image segmentation and classification.

Designed supervised learning and deep feature ensemble methods for diagnosing pneumonia.

EDUCATIONAL DEGREES:

Industrial PhD in Biomedical Engineering (Marie Skłodowska-Curie Actions Fellow)

Politecnico di Torino, Italy and GPI S.p.A., Trento, Italy May 2026

Dissertation Title: AI-Driven, Interpretable Pattern Recognition for Multimodal Medical Data in Clinical Applications

MS in Computer Science

COMSATS University, Islamabad, Wah Campus (Pakistan) July 2021

Dissertation Title: Deep Features Ensemble Technique to Diagnose Bacterial and Viral Pneumonia Based on Supervised Learning

BS in Computer Science (Honours) Muhammad Nawaz Sharif University of Agriculture, Multan (Pakistan) Sept 2019

Project Title: An Automated Content-Based Facial Image Retrieval using Convolutional Neural Networks

RESEARCH INTERESTS:

Artificial Intelligence and Machine Learning

Explainable AI and Trustworthy AI Systems

Computer Vision and Biomedical Image Analysis

Natural Language Processing and Vision-Language AI for Medical Applications

Multimodal Learning and Multi-Agent AI Systems

Speech, Voice, and Audio Signal Analysis

AI for Digital Health and Clinical Decision Support

TECHNICAL SKILLS:

Tools and Frameworks (Matlab ®, Tensorflow, Keras, PyTorch, Caffe, OpenCV, Pandas and MS VS ®)

Languages (R, Python, Bash, C/C++, C#, JAVA)

Web Development (Django & Flask under Python, PHP, HTML, CSS, JavaScript, React, and React Native)

Devops (Docker, Jenkins)

Graphical tools (Open-GL and Blender)

Databases (MySQL, MS SQL, MongoDB)

Version Control (Git, GitHub, and GitLab)

Project Management & Collaboration Tools (Jira, Confluence, Notion, Slack, Microsoft Teams, MS Project, LaTeX, Overleaf)

PROFESSIONAL TRAINING COURSES:

Deep Learning (Python) for Neuroscience EEG Practical course, Udemy (Aug 2025)

Worker Occupational Safety Training Course, VEGA Formazione with GPI SpA (Oct 2024)

Workers' Specific Safety Training - Low Risk, VEGA Formazione with GPI SpA (Oct 2024)

RESEARCH FUNDING & FELLOWSHIPS:

TALiA (B69J24001180005)- Ministry of Economy and Finance, Trento, Italy (Mar 2025 - Present)

PARENT (956394)-Marie Skłodowska-Curie-Innovative Training Network 2020 (Aug 2022 - March 2025)

Achieved Punjab Educational Endowment Fund (PEEF) and educational scholarship from Fauji Foundation for higher and undergraduate studies, Pakistan (Jun 2011 - Aug 2019)

JOURNAL PUBLICATIONS:

Syed Baqir Hussain Shah, Farwa Naseer, Syed Adil Hussain Shah, Kashif Razzaq, Tahir Javed, Qandeel Asghar, Gohar Bano Zaidi, Giacomo Di Benedetto, Syed Taimoor Hussain Shah, Syed Bilal Hussain, Marco Agostino Deriu (2026). B2-GraftingNet: A Hybrid Deep-Machine Learning Framework with Explainable AI for Automated Grape Leaf Disease Detection. ICCK Journal of Image Analysis and Processing, 2(1). DOI: <https://doi.org/10.62762/JIAP.2026.937901>.

Angel Ruiz-Zafra, Janet Pigueiras-del-Real, Jose Heredia-Jimenez, Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, & Lionel C. Gontard (2025). PyBodyTrack: A python library for multi-algorithm motion quantification and tracking in videos. SoftwareX, 31, 102272. <https://doi.org/10.1016/j.softx.2025.102272>.

Riccardo Sartori, Francesca Marinaro, Francesco Tommasi, Andrea Buccoliero, Mattia Zene, Syed Adil Hussain Shah, Andrea Ceschi (2025). A scoping review on the use of voice biomarkers for emotional assessment. European Journal of Psychological Assessment. DOI: <https://doi.org/10.1027/1015-5759/a000915>.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Andrea Buccoliero, Iqra Iqbal Khan, Syed Baqir Hussain Shah, Angelo Di Terlizzi, & Giacomo Di Benedetto (2025). Analyzing neonatal vocal expression: Methodological approaches to identifying neurological and psychiatric signatures. Journal of Multiscale Neuroscience, 4(2): 158-176. DOI: <https://doi.org/10.56280/1703023560>.

Janet Pigueiras-Del-Real, Lionel C. Gontard, Isabel Benavente-Fernández, Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Simón P. Lubián-López, Ángel Ruiz-Zafra (2025). Evaluating Methods for Reliable Non-Invasive Motion Quantification as an Indicator of Favorable Neonatal Progression. Healthcare (MDPI). DOI: <https://doi.org/10.3390/healthcare13131577>.

Syed Adil Hussain Shah, Syed Taimoor Hussain Shah, Abdul Muiz Fayyaz, Syed Baqir Hussain Shah, Mussarat Yasmin, Mudassar Raza, Angelo Di Terlizzi, Marco Agostino Deriu (2025). Improving biomedical image pattern identification by Deep B4-GraftingNet: application to pneumonia detection. IET Image Processing. <https://doi.org/10.1049/ipr2.70064>.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Konstantinos Panagiotopoulos, Janet Pigueiras-del-Real, Kainat Qayyum, Syed Baqir Hussain Shah, Shahzad Ahmad Qureshi, Angelo Di Terlizzi, Giacomo Di Benedetto, Marco Agostino Deriu (2025). Artificial intelligence coupled with the Internet of Things targeting neurodevelopmental challenges in preterm neonates. *Journal of Multiscale Neuroscience* 4(1), 32-56. <https://doi.org/10.56280/1680389540>.

Shahzad Ahmad Qureshi, Aziz-UI-Rehman, Lal Hussain, Touseef Sadiq, Syed Taimoor Hussain Shah, Adil Aslam Mir, Muhammad Amin Nadim, Darnell K. Adrian Williams, Tim Q Duong, Qurat-UI-Ain Chaudhary, Natasha Habib, Asrar Ahmad, Syed Adil Hussain Shah (2024). Breast cancer detection using mammography: Image processing to deep learning. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2024.3523745>.

Syed Adil Hussain Shah, Syed Taimoor Hussain Shah, Andrea Buccoliero, Syed Baqir Hussain Shah, Angelo Di Terlizzi, Giacomo Di Benedetto, Marco Agostino Deriu (2024). Explainable AI-based skin cancer detection using CNN, particle swarm optimization, and machine learning. *Journal of Imaging*, 10(12), 332. <https://doi.org/10.3390/jimaging10120332>.

Janet Pigueiras-del-Real, Angel Ruiz-Zafra, Isabel Benavente-Fernández, Simón P Lubián-López, Syed Adil Hussain Shah, Syed Taimoor Hussain Shah, Lionel C Gontard (2024). NeoVault: empowering neonatal research through a neonate data hub. *BMC Pediatrics*, 24(1), 1-9. <https://doi.org/10.1186/s12887-024-05276-y>.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Iqra Iqbal Khan, Atif Imran, Syed Baqir Hussain Shah, Atif Mehmood, Shahzad Ahmad Qureshi, Mudassar Raza, Angelo Di Terlizzi, Marco Cavaglià, Marco Agostino Deriu (2024). Data-driven classification and explainable-AI in the field of lung imaging. *Frontiers in Big Data*. <https://doi.org/10.3389/fdata.2024.1393758>.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Shahzad Ahmad Qureshi, Angelo Di Terlizzi, Marco Agostino Deriu (2023). Automated facial characterization and image retrieval by convolutional neural networks. *Frontiers in Artificial Intelligence*. <https://doi.org/10.3389/frai.2023.1230383>.

Syed Taimoor Hussain Shah, Shahzad Ahmad Qureshi, Aziz ul Rehman, Syed Adil Hussain Shah, Arslan Amjad, Adil Aslam Mir, Amal Alqahtani, David A Bradley, Mayeen Uddin Khandaker, Mohammad Rashed Iqbal Faruque, Muhammad Rafique (2021). A novel hybrid learning system using modified breaking ties algorithm and multinomial logistic regression for classification and segmentation of hyperspectral images. *Applied Sciences*, 11(16). <https://doi.org/10.3390/app11167614>.

CONFERENCE PUBLICATIONS:

Syed Baqir Hussain Shah, Musfarah Wajid, Syed Adil Hussain Shah, Areeba Khan, Gohar Bano Zaidi, Andrea Buccoliero, Waheed Ahmad Khan, Syed Taimoor Hussain Shah, Marco Agostino Deriu (2025). Explainable Deep Learning for Shoulder Fracture Detection: EfficientNet-Based Evaluation on the MURA Dataset. *ICECET 2026 (6th International Conference on Electrical, Computer and Energy Technologies)*. Track: Artificial Intelligence & Deep Learning. Manuscript ID: 702 (Accepted).

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Syed Baqir Hussain Shah, Janet Pigueiras-del-Real, Iqra Iqbal Khan, Ammara Zamir, Qandeel Asghar, Shahzad Ahmad Qureshi, Angel Ruiz-Zafra, Lionel Cervera Gontard, and Marco Agostino Deriu (2025). Multiclass Lung Cancer Detection Using Deep Learning and Explainable AI on Chest CT Scans. *22nd IBCAST 2025 - Section: Artificial Intelligence & Software Technologies*. Manuscript ID: 134 (Blind Review & 18% Acceptance Rate - Presented).

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Konstantinos Panagiotopoulos, Janet Pigueiras-del-Real, Kainat Qayyum, Syed Baqir Hussain Shah, Andrea Buccoliero, Angelo Di Terlizzi, and Marco Agostino Deriu (2025). Explainable Emotion Recognition Using Xception-Based Feature Extraction and Supervised Machine Learning on the RAVDESS Dataset. 2025 IEEE Medical Measurements & Applications (MeMeA). DOI: <https://doi.org/10.1109/MeMeA65319.2025.11068008>.

Syed Adil Hussain Shah, Angelo Di Terlizzi, Marco Agostino Deriu (2022). Intelligent system development to monitor the neonatal behavior: A review. DETERMINED, 133-146. <https://doi.org/10.5281/zenodo.8009881>.

Syed Taimoor Hussain Shah, Shahzad Ahmad Qureshi, Aziz ul Rehman, Syed Adil Hussain Shah, Jamal Hussain (2020). Classification and Segmentation Models for Hyperspectral Imaging - An Overview. In Intelligent Technologies and Applications. INTAP (Springer). https://doi.org/10.1007/978-3-030-71711-7_1.

Syed Taimoor Hussain Shah, Syed Gibran Javed, Abdul Majid, Syed Adil Hussain Shah, Shahzad Ahmad Qureshi (2019). Novel Classification Technique for Hyperspectral Imaging using Multinomial Logistic Regression and Morphological Profiles with Composite Kernels. In Proceedings of the 16th International Bhurban Conference on Applied Sciences and Technology (IBCAST). IEEE. <https://doi.org/10.1109/IBCAST.2019.8667162>.

PREPRINTS:

Syed Baqir Hussain Shah, Farwa Naseer, Kashif Razzaq, Syed Adil Hussain Shah, Tahir Javaid, Qandeel Asghar, Giacomo Di Benedetto, Syed Bilal Hussain, Syed Taimoor Hussain Shah, and Marco Agostino Deriu (2025). B2-GraftingNet: A Hybrid Deep-Machine Learning Framework with Explainable AI for Automated Grape Leaf Disease Detection. Preprints. DOI: <https://www.preprints.org/manuscript/202512.1054>.

Shahzad Ahmad Qureshi, Muhammad Abdul Basit, Qasim Ijaz, Syed Taimoor Hussain Shah, Syed Mesam Tamar Kazmi, Inam ul Haq, Ghulam Mustafa, Syed Adil Hussain Shah (2025). Bias-Resilient Ensemble of Transfer Learning Models for Automated COVID-19 Detection from Chest Radiographs. Preprints, 202508.1553. DOI: <https://doi.org/10.20944/preprints202508.1553.v1>.

Shahzad Ahmad Qureshi, Muhammad Owais, Ayesha Hakim, Qasim Ijaz, Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Adil Aslam Mir (2025). VRTMS: A 3D-Virtual Reality Training and Maintenance System as a Modular Cognitive Approach to a Prototype Infrared Computed Tomography Machine Assembly. Preprints. DOI: <https://doi.org/10.20944/preprints202505.0899.v1>.

Shahzad Ahmad Qureshi, Aziz-UI-Rehman, Lal Hussain, Touseef Sadiq, Syed Taimoor Hussain Shah, Adil Aslam Mir, Darnell K. Adrian Williams, Tim Q Duong, Qurat-UI-Ain Chaudhary, Natasha Habib, Asrar Ahmad, Syed Adil Hussain Shah (2024). Breast cancer detection using mammography: Image processing to deep learning. Preprints. <https://doi.org/10.20944/preprints202405.0527.v1>.

SOFTWARE, DATA & OPEN SCIENCE:

Syed Baqir Hussain Shah, Farwa Naseer, Kashif Razzaq, Syed Adil Hussain Shah, Tahir Javaid, Qandeel Asghar, Giacomo Di Benedetto, Syed Bilal Hussain, Syed Taimoor Hussain Shah, and Marco Agostino Deriu (2025). Web Platform for B2-GraftingNet: A Hybrid Deep-Machine Learning Framework with Explainable AI for Automated Grape Leaf Disease Detection. Zenodo. DOI: <https://doi.org/10.5281/zenodo.17353656>.

Angel Ruiz-Zafra, Janet Pigueiras-del-Real, Jose Heredia-Jimenez, Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Lionel C. Gontard. (2025). PyBodyTrack: A python library for multi-algorithm motion quantification and tracking in videos. *SoftwareX*, 31, 102272. <https://doi.org/10.1016/j.softx.2025.102272>.

GitHub: <https://github.com/bihut/PyBodyTrack.git>.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Ammara Zamir, Kainat Qayyum, Syed Baqir Hussain Shah, Syeda Maryam Fatima, Marco Agostino Deriu (2025). Mediapipe-based Preprocessed VGGFace2 Dataset. Zenodo. <https://doi.org/10.5281/zenodo.15078556>.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Konstantinos Panagiotopoulos, Janet Pigueiras-del-Real, Syed Baqir Hussain Shah, Marco Agostino Deriu (2025). Xception-based Facial Expression Recognition with SHAP-Driven Explainability. Zenodo. DOI: <https://doi.org/10.5281/zenodo.14809940>.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Syed Baqir Hussain Shah, Shahzad Ahmad Qureshi, Angelo Di Terlizzi, Marco Agostino Deriu. (2023). Code of Automated Facial Characterization and Image Retrieval by Convolutional Neural Networks. Zenodo. DOI: <https://doi.org/10.5281/zenodo.14865792>.

UNDER-REVIEW PUBLICATIONS:

Syed Adil Hussain Shah, Syed Taimoor Hussain Shah, Janet Pigueiras del Real, Francesca Marinaro, Andrea Buccoliero, Syed Baqir Hussain Shah, Angelo Di Terlizzi, Shahzad Ahmad Qureshi, Angel Ruiz Zafra, Lionel Cervera Gontard, and Marco Agostino Deriu. VocalXpressNet: A Modified Xception Network with Explainable AI for Identification of Vocal Emotions and Infant Pain Levels. *Multimedia Tools and Applications - Springer Nature*. Manuscript ID: MTAP-D-25-04127R1 (Under Revision).

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Silvia Godio, Karim Kassem, Shahzad Ahmad Qureshi, Syed Bilal Hussain, and Marco Agostino Deriu. Explainable AI-Based Two-Stage Swin Transformer for Grapevine Leaf-Based Variety and Disease Classification. *Information Processing in Agriculture*. Manuscript ID: IPA-D-26-00347 (Under Revision).

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Syed Baqir Hussain Shah, Janet Pigueiras-del-Real, Roa'a Khaled, Ammara Zamir, Syed Muhammad Jawwad Riaz, Shahzad Ahmad Qureshi, Angel Ruiz-Zafra, Lionel C. Gontard, Marco Agostino Deriu. FAITHAI: A GAN-Driven and Explainable AI CNN-Based Approach for Infant Disease Detection via Facial Analysis. *Biomedical Signal Processing and Control*. Manuscript ID: BSPC-D-26-07653.

Syed Taimoor Hussain Shah, Syed Adil Hussain Shah, Janet Pigueiras-del-Real, Konstantinos Panagiotopoulos, Syed Baqir Hussain Shah, Angelo Di Terlizzi, Isabel Benavente-Fernández, Simón P. Lubián-López, Marco Agostino Deriu, Lionel C. Gontard, Angel Ruiz-Zafra. Geometry-and Dynamics-Aware Imputation and Generation of Neonatal Pose Sequences: Linear, Angular, Inertia-Based, and LSTM Models.

Frontiers in Big Data. Manuscript ID: 1872138.

Syed Baqir Hussain Shah, Musfarah Wajid, Syed Adil Hussain Shah, Silvia Godio, Karim Kassem, Gohar Bano Zaidi, Shahzad Ahmad Qureshi, Syed Taimoor Hussain Shah, Marco Agostino Deriu. Explainable Two-Stage Xception-Swin Transformer Learning for Body-Part-Aware Fracture Detection in Musculoskeletal X-Rays. *Journal of Imaging*. Manuscript ID: jimaging-4244005.

PARTICIPATION IN RESEARCH PROJECTS:

CONFERENCE & SEMINARS:

ACADEMIC SERVICE:

Journal Reviewer

Nature (Scientific Reports); PLOS One; Wiley (Applied Computational Intelligence and Soft Computing); Frontiers (Frontiers in Psychology, Frontiers in Medicine, Frontiers in Cardiovascular Medicine); IEEE (Journal of Biomedical and Health Informatics)

TEACHING EXPERIENCE:

Guest Lecturer

Università degli studi di Verona, Italy 07 May 2025

Digital Health Technologies (Master's level)

Università degli studi di Verona, Italy 08 May 2026

Agentic AI for Healthcare Innovation (Master's level)

SUPERVISION & MENTORSHIP:

PUBLIC ENGAGEMENT & OUTREACH:

2022 INNOVALAB GPI Event (13-14 Septemeber 2022)

2022 European Researchers' Night (30 September 2022)

2023 European Researchers' Night (30 September 2023)

Interim Inspection Visit - Talia Project event (11 May 2026)

REFERENCES:
